Functional Annotation Report: *pucL* / PP_4287 (UniProt Q88F12) in *Pseudomonas putida* KT2440

1. Summary (Answer to the Research Question)

The gene **pucL** (ordered locus **PP_4287**; UniProt **Q88F12**) of *Pseudomonas putida* KT2440 encodes **2-oxo-4-hydroxy-4-carboxy-5-ureidoimidazoline (OHCU) decarboxylase (EC 4.1.1.97)**, a small (171 aa) soluble, cofactor-independent enzyme. It catalyzes the **third and final step of the uricase-dependent purine/urate degradation pathway**, converting the unstable intermediate OHCU into **(S)-allantoin + CO_2**. Its function is to guarantee **rapid and stereospecific** production of the (S)-enantiomer of allantoin — the biologically relevant form used by downstream ureide-catabolizing enzymes — rather than the slow, racemic product formed by spontaneous non-enzymatic decay. The enzyme acts in the **cytoplasm**, integrated into purine catabolism that ultimately allows bacteria to use purines/allantoin as a nitrogen (and sometimes carbon) source.

> **Identity verification (per task requirement):** UniProt Q88F12 is annotated `2-oxo-4-hydroxy-4-carboxy-5-ureidoimidazoline decarboxylase`, EC 4.1.1.97, gene `pucL`/`PP_4287`, organism *Pseudomonas putida* KT2440, with the diagnostic **OHCU_decarboxylase** domain (Pfam **PF09349**; InterPro IPR018020, IPR017580, IPR036778). The catalytic-activity and pathway annotations retrieved directly from UniProt match the protein description exactly. **Caveat on the symbol:** the label *pucL* elsewhere (notably *Bacillus subtilis*) denotes **urate oxidase (uricase)**; in this *P. putida* genome annotation *pucL* is simply the assigned locus symbol, and the domain/EC evidence unambiguously identifies the product as an **OHCU decarboxylase**, not a uricase. All functional literature below concerns the OHCU decarboxylase (URAD/UraD) family, which is the correct protein.

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2. Molecular Identity and Sequence Features

Property	Value	Source
UniProt accession	Q88F12 (Q88F12_PSEPK)	UniProt
Gene / locus	*pucL* / PP_4287	UniProt / EMBL AAN69867.1
Length	171 aa (soluble, α-helical; no signal peptide/TM segment)	UniProt sequence
EC number	4.1.1.97	UniProt
Reaction	5-hydroxy-2-oxo-4-ureido-2,5-dihydro-1H-imidazole-5-carboxylate (OHCU) + H^+ → (S)-allantoin + CO_2	UniProt catalytic activity
Pathway	Purine metabolism; urate degradation; (S)-allantoin from urate: **step 3/3** (UniPathway UPA00394)	UniProt
Domain / family	OHCU_decarboxylase (Pfam PF09349; InterPro IPR018020/IPR017580/IPR036778); eggNOG **COG3195**	UniProt/ InterPro
Evidence level	Protein existence: **4 – Predicted** (function inferred from homology/ annotation; no direct experimental study of this specific protein)	UniProt

3. Primary Function: The Catalyzed Reaction and Substrate Specificity

The **uricase (urate-oxidase) degradation pathway** converts the poorly soluble purine end-product uric acid to the much more soluble ureide **allantoin** in three enzymatic steps [Ramazzina et al., 2006, P 16462750]:

1. **Urate oxidase (uricase)** — urate → **5-hydroxyisourate (HIU)**
2. **HIU hydrolase (HIUase / transthyretin-related)** — HIU → **OHCU**
3. **OHCU decarboxylase (this protein, pucL/PP_4287) — OHCU → (S)-allantoin** + **CO_2**

Ramazzina and colleagues established this three-enzyme scheme by phylogenetic genome comparison, showing that the two genes downstream of uricase (HIUase and OHCU decarboxylase) "catalyze two consecutive steps following urate oxidation to 5-hydroxyisourate (HIU): hydrolysis of HIU to give 2-oxo-4-hydroxy-4-carboxy-5-ureidoimidazoline (OHCU) and **decarboxylation of OHCU to give S-(+)-allantoin**" [PMID 16462750].

Substrate specificity: The enzyme is highly specialized for its single natural substrate, OHCU. Its product complex with **(S)-allantoin** was captured crystallographically [Kim, Park & Rhee 2007, P 17567580], and a structural isomer of the purine hypoxanthine, **allopurinol**, acts as a **competitive inhibitor** of the bacterial enzyme (K_i = 30 ± 2 μM in *Klebsiella pneumoniae*), consistent with an active site tuned to the imidazoline/purine-derived scaffold [French & Ealick 2010, P 20826786].

4. Physiological Rationale: Why the Cell Needs This Enzyme

OHCU is an **unstable intermediate** that spontaneously decarboxylates. Without the enzyme, the reaction still proceeds but is slow and **non-stereospecific**: "Urate oxidation produces **racemic allantoin on a time scale of hours**, whereas the full enzymatic complement produces **dextrorotatory allantoin on a time scale of seconds**" [Ramazzina et al., 2006, P 16462750].

Thus pucL/OHCU decarboxylase performs two coupled jobs: - **Kinetic acceleration** — converting OHCU to allantoin in seconds rather than hours. - **Stereochemical control** — producing exclusively **(S)-(+)-allantoin**, the enantiomer recognized by downstream ureide-pathway enzymes (allantoinase, allantoate amidohydrolase, etc.).

This resolves the long-standing question of how organisms selectively make (S)-allantoin [Kim et al., 2007, P 17567580].

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5. Structural Basis and Catalytic Mechanism

Crystallographic and mutagenesis studies of family members (zebrafish, *Klebsiella pneumoniae*, human) define the mechanism that Q88F12 is expected to share (Pfam PF09349):

- **Fold / oligomer:** a **homodimeric, all-α-helical protein of a novel structural motif** [Kim et al., 2007, P 17567580].
- **Cofactor independence:** "the active site in each monomer contains **no cofactors**, distinguishing this enzyme mechanistically from other cofactor-dependent decarboxylases" [PMID 17567580]. Catalysis is direct, with no metal ion, PLP, or thiamine cofactor.
- **Catalytic residue / stereocontrol:** "the **invariant histidine** residue in the OHCU decarboxylase family plays an essential role in producing (S)-allantoin through a **proton transfer from the hydroxyl group at C4 to C5 at the re-face of OHCU**" [PMID 17567580].
- **Induced-fit organization:** apo vs. allantoin-bound structures of the *K. pneumoniae* enzyme show that "**ligand binding organizes the active site residues for catalysis**"; the inhibitor allopurinol disrupts this organization (confirmed by circular dichroism) [French & Ealick

2010,

P 20826786].

The 171-aa length, all-helical prediction, and conserved OHCU_decarboxylase domain of Q88F12 are fully consistent with this family architecture.

Sequence-based confirmation of the catalytic residue (this work): A Needleman–Wunsch alignment of Q88F12 against reviewed *standalone* OHCU decarboxylase orthologs — human URAD (A6NGE7), zebrafish (A1L259), and mouse (Q283N4) — shows the family's single conserved histidine mapping to **His73 of Q88F12** in all three orthologs (~30–35% overall identity). His73 thus corresponds to the invariant active-site histidine that Kim et al. (2007) demonstrated is essential for the proton transfer generating (S)-allantoin, giving evolutionary/sequence evidence that Q88F12 is catalytically competent.

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6. Subcellular Localization

No experimental localization data exist for Q88F12 (UniProt evidence level "Predicted"). However, multiple lines of inference place it in the **cytoplasm**: - The protein has **no signal peptide and no transmembrane segment** (soluble 171-aa helical protein). - Its substrate OHCU is generated intracellularly by soluble upstream enzymes (uricase and HIU hydrolase), so the enzyme must act where these intermediates are produced. - Family members characterized biochemically are soluble cytosolic proteins.

Conclusion: cytoplasmic (soluble), acting in a metabolon-like sequence with uricase and HIU hydrolase.

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7. Pathway Context and Broader Biological Role

OHCU decarboxylase sits at the end of the **purine catabolism → ureide** route. In many bacteria this pathway enables purines/urate/allantoin to be used as **nitrogen (and sometimes carbon) sources**:

- In *Klebsiella pneumoniae*, "purines can be used as the sole source of nitrogen ... under aerobic conditions," with allantoin further degraded by allantoinase and allantoate amidohydrolase, under nitrogen-responsive regulation (NtrC/NAC and a GntR-family repressor HpxS) [Guzmán et al., 2011, P 21357483].

- In *Streptomyces coelicolor*, allantoin catabolic genes are controlled by the repressor **AllR**, responding to allantoic acid/glyoxylate [Navone et al., 2015, P 26187964], underscoring that this branch of metabolism is tightly regulated by nutrient availability.
- The evolutionary logic of the pathway is highlighted by its **loss in hominids**: humans retain a pseudogenized/inactivated set (UOX, URAHP, URAD), and recombinant human OHCU decarboxylase is a well-folded but catalytically impaired enzyme, reflecting relaxed selection after uricase loss [Rodrigues et al., 2026, P 41956358]. This confirms URAD/pucL as an evolutionarily conserved, functional enzyme in organisms (like *P. putida*) that retain the complete pathway.

Genomic-context evidence in *P. putida* KT2440 (this work): *pucL*/PP_4287 lies within a contiguous, same-strand **purine/urate catabolic gene cluster** (KEGG genome, ~4.877–4.879 Mb):

Locus	Product	KO / EC
PP_4284	adenine/guanine/hypoxanthine permease	K06901
PP_4285	**5-hydroxyisourate (HIU) hydrolase** (pathway step 2, makes OHCU)	K07127 / EC 3.5.2.17
PP_4286	allantoinase	K16842 / EC 3.5.2.5
PP_4287	**OHCU decarboxylase (pucL, step 3)**	K13485 / EC 4.1.1.97
PP_4288	ureidoglycolate lyase	K01483 / EC 4.3.2.3
PP_4290	uric acid permease	K24206

The enzyme that generates *pucL*'s substrate (HIU hydrolase, PP_4285) is immediately adjacent, and downstream allantoin-degrading enzymes (allantoinase, ureidoglycolate lyase) and purine/urate transporters are clustered together. This "guilt-by-association" organization independently confirms that PP_4287 operates *in situ* within a coordinated **urate → allantoin → glyoxylate/ammonia** degradation route.

In *P. putida* KT2440 — a soil saprophyte that thrives on diverse organic substrates — this cluster contributes to **purine/urate scavenging for nitrogen recycling**, converting urate into assimilable allantoin and beyond.

Pathway-completeness note (this work): KEGG KO-link queries confirm KT2440 encodes the entire *downstream* module — HIU hydrolase (PP_4285), OHCU decarboxylase (PP_4287), allantoinase (PP_4286), ureidoglycolate lyase (PP_4288) — but **no canonical urate oxidase** is annotated (neither K00365 uricase, K16838 HpxO, nor K13484 PuuD-type). Because HIU hydrolase acts specifically on 5-hydroxyisourate (the direct product of urate oxidation), an upstream urate-oxidizing activity must nonetheless exist, implying a **divergent/unannotated urate oxidase** in KT2440. This does not change the assignment of PP_4287 as the step-3 OHCU decarboxylase, but flags the first step as annotation-incomplete and a target for experimental verification.

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8. Supported and Refuted Hypotheses

Supported: - H1 — Q88F12/pucL is an **OHCU decarboxylase (EC 4.1.1.97)** catalyzing OHCU → (S)-allantoin + CO_2. *(UniProt annotation + domain + family literature.)* - H2 — It performs **step 3/3** of the urate→(S)-allantoin pathway and enforces **fast, stereospecific** (S)-allantoin formation.

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P *16462750*

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P *17567580*

- H3 — The enzyme is **cofactor-independent**, a **helical homodimer**, using an **invariant histidine** for stereocontrol.

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P *17567580*

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P *20826786*

- H4 — It functions in the **cytoplasm**, embedded in nitrogen-yielding purine catabolism.

(Sequence inference + pathway context, 21357483*.)*

- H5 — In *P. putida* KT2440, PP_4287 is embedded in a **physically clustered purine/urate–allantoin catabolic operon** (HIU hydrolase, allantoinase, ureidoglycolate lyase, purine/urate permeases). *(KEGG genomic-context analysis, this work.)* - H6 — Q88F12 retains the **invariant catalytic His73** of the family. *(Ortholog alignment, this work; mechanism per* 17567580*.)*

Refuted / excluded: - The symbol *pucL* does **NOT** here indicate **urate oxidase/uricase** (its meaning in *B. subtilis*); the EC number, catalytic reaction, and Pfam PF09349 domain exclude a uricase assignment.

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9. Limitations and Future Directions

- **Direct experimental evidence is lacking for the specific *P. putida* protein** (UniProt "Predicted"). All mechanistic and structural detail derives from orthologs (mouse, zebrafish, *K. pneumoniae*, human) and from robust domain/pathway conservation.
- No experimental subcellular localization or in-vitro kinetics for Q88F12 itself; these are inferred.
- **Upstream step not annotated:** KT2440 lacks a KEGG-annotated classical urate oxidase, so the identity of the enzyme producing 5-hydroxyisourate (the substrate source for the module) remains to be established experimentally.
- **Future work:** recombinant expression and steady-state kinetics of Q88F12 with OHCU; confirmation of the His73 catalytic residue by site-directed mutagenesis; identification of the divergent urate oxidase feeding PP_4285/PP_4287; growth assays testing urate/allantoin as sole nitrogen sources in KT2440 and its regulation; and an experimental (or AlphaFold-validated) structure to confirm the homodimeric helical fold.

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10. Key References

- Ramazzina et al. *Nat. Chem. Biol.* 2006 — pathway completion; racemic vs. (S)-allantoin kinetics.

  P 16462750

- Kim, Park & Rhee 2007 — OHCU decarboxylase–(S)-allantoin structure; cofactor-free mechanism; invariant His.

  P 17567580

- French & Ealick 2010 — *K. pneumoniae* OHCU decarboxylase structures, kinetics, allopurinol inhibition.

  P 20826786

- Rodrigues et al. 2026 — human OHCU decarboxylase (URAD) characterization; evolutionary conservation.

  P 41956358

- Guzmán et al. 2011 — purines as nitrogen source; allantoin pathway regulation in *K. pneumoniae*.

  P 21357483

- Navone et al. 2015 — AllR regulation of allantoin pathway in *S. coelicolor*.

  P 26187964